Investigating Temperature & Wind Speed

Exam code: 0610 & 0970

Download PDF

Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
Updated on 13 January 2026

SaveMyExams

Did this video help you? Yes No

Help me understand

Investigating the Effect of Temperature & Wind Speed on Transpiration Rate

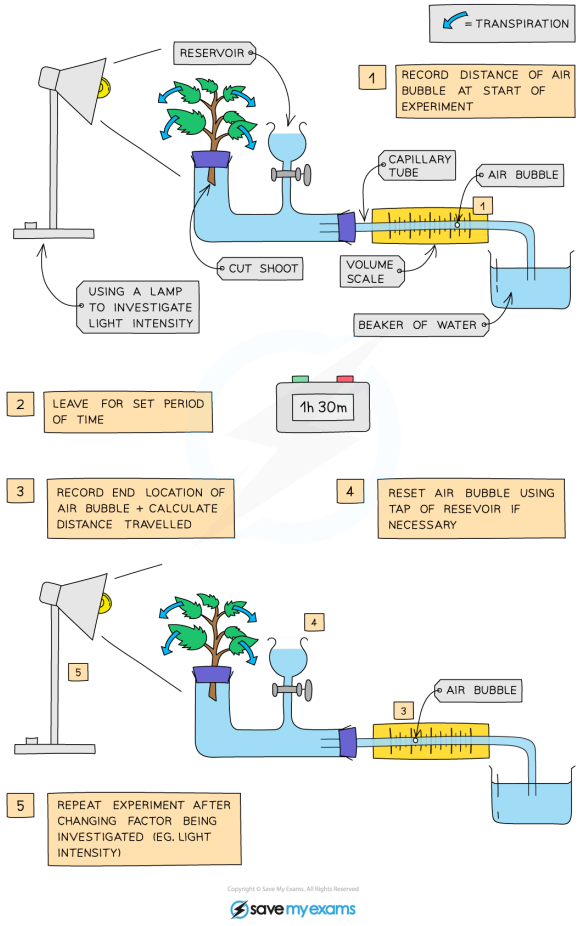
Method:

- Cut a shoot underwater to **prevent air entering the xylem** and place in tube
- Set up the apparatus as shown in the diagram and make sure it is airtight, using vaseline to seal any gaps
- Dry the leaves of the shoot (wet leaves will affect the results)
- Remove the capillary tube from the beaker of water to allow a single **air bubble** to form and place the tube back into the water
- Set up the environmental factor you are investigating
- Allow the plant to adapt to the new environment for 5 minutes
- Record the **starting location** of the air bubble
- Leave for a set period of time
- Record the **end location** of air bubble
- Change the wind speed or temperature (only one - whichever factor is being investigated)
- Reset the bubble by opening the tap below the reservoir
- Repeat the experiment

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



An experimental setup for testing the effect of light intensity on transpiration rates. The apparatus can be modified to test the effects of temperature and wind speed.

- Environmental factors can be investigated in the following ways:
  - **Temperature**: Temperature of room (cold room and warm room)
    - As temperature increases, the rate of transpiration also increases
  - **Wind speed**: Use an electric fan to mimic different wind speeds
    - As wind speed increases, the rate of transpiration also increases

Examiner Tips and Tricks

Remember when designing an investigation to ensure a fair test you must keep all factors the same other than the one you are investigating.

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself Flashcards

Was this revision note helpful? Yes No

Previous Next

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.